

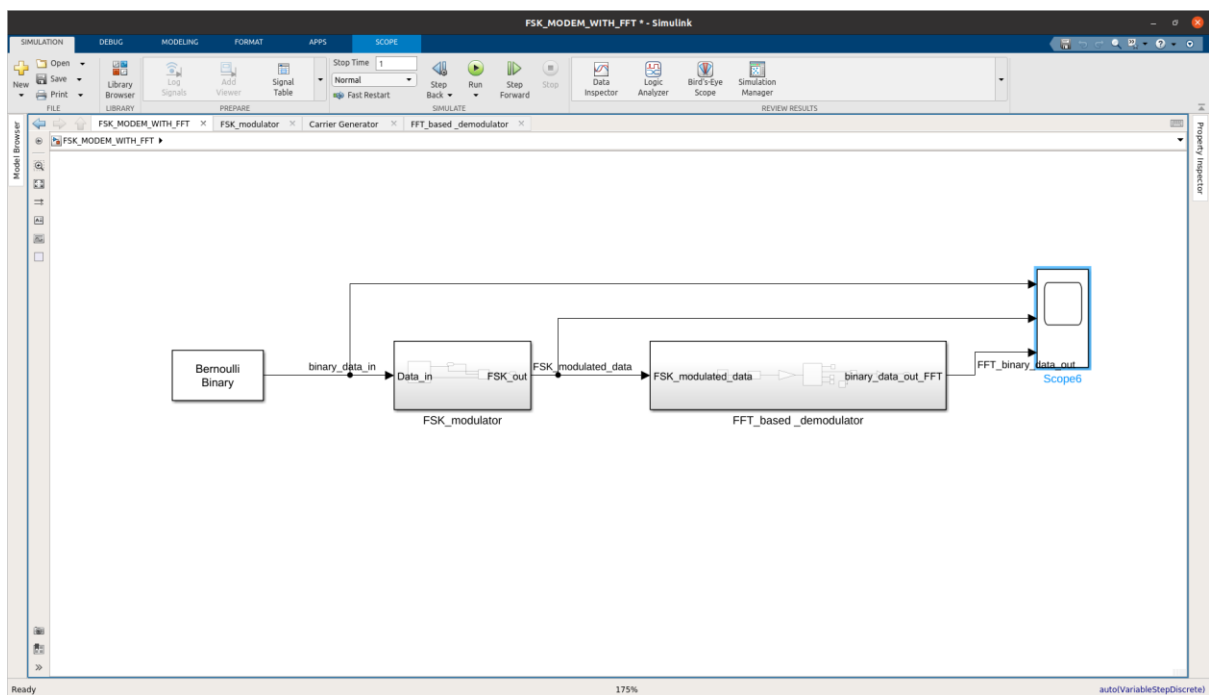
INTRODUCTION

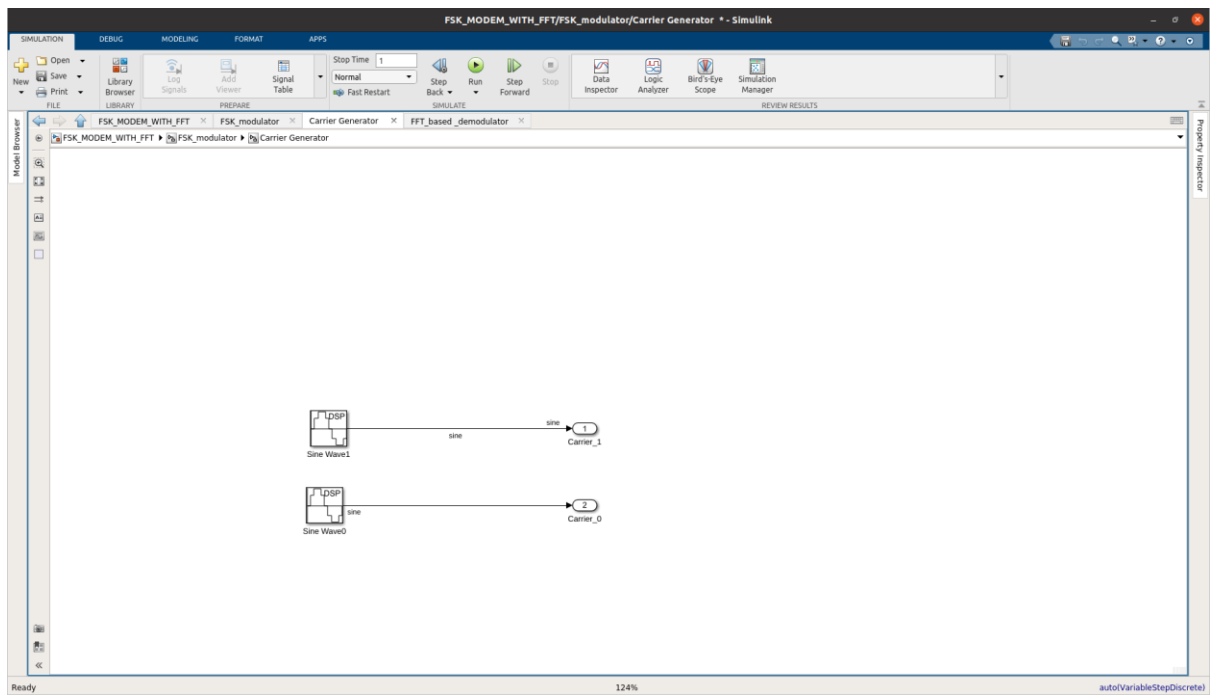
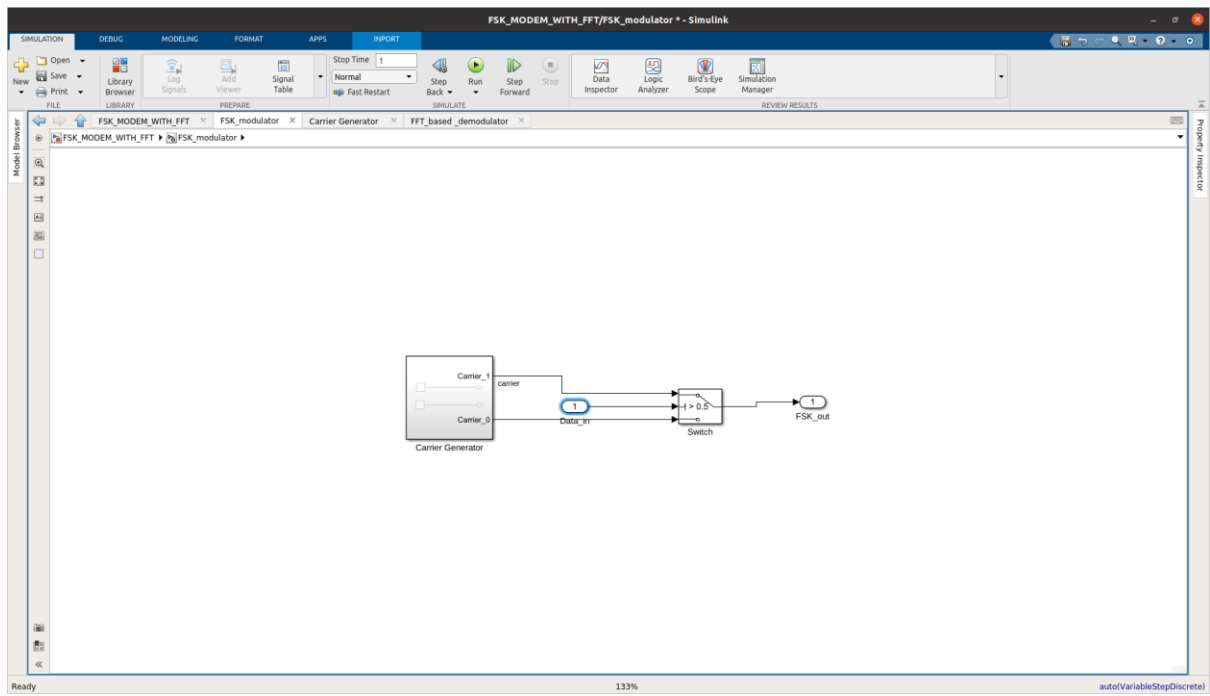
Frequency Shift Keying (FSK) is a digital modulation technique in which binary data is transmitted by shifting the frequency of a carrier signal between two distinct values. In binary FSK (BFSK), one frequency represents a logical '0' and another represents a logical '1'. To recover the transmitted data, the receiver must determine which frequency is present during each symbol period. One effective way to do this is by using the Fast Fourier Transform (FFT), which converts time-domain signal segments into their frequency-domain representation, allowing easy identification of the dominant frequency in each symbol and thus the corresponding bit.

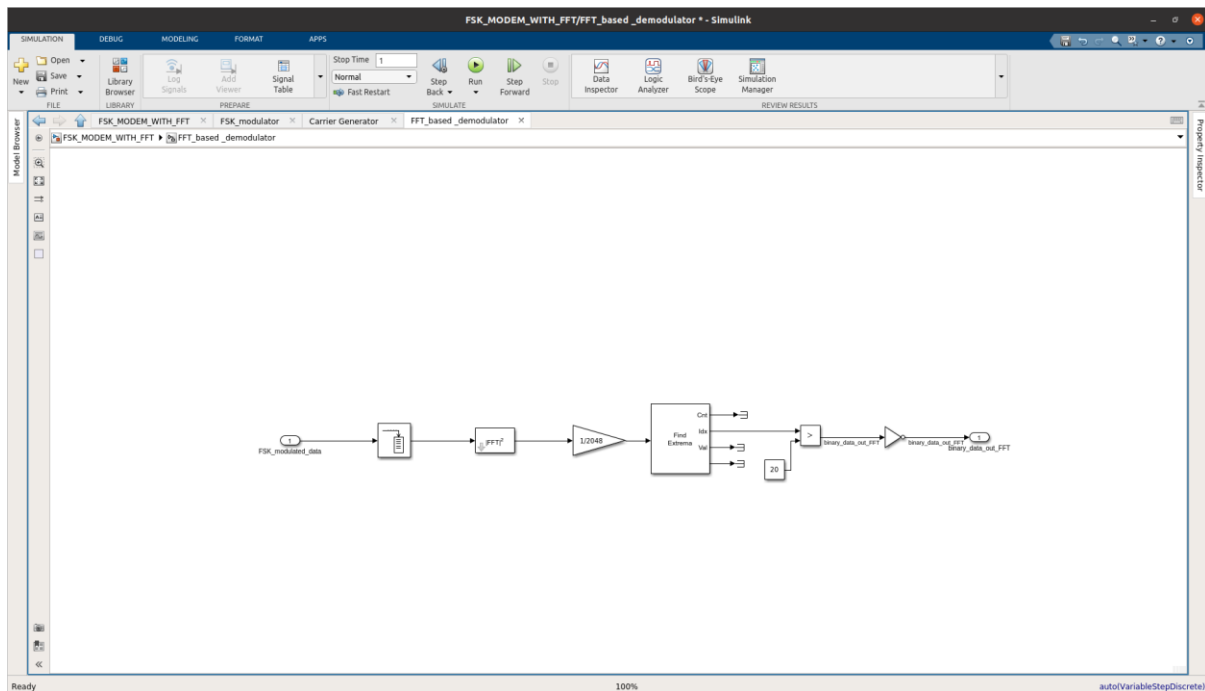
SIMULINK FFT BASED FSK DEMODULATION

This model implements a **binary FSK demodulator** using FFT.

It analyzes the spectrum of each symbol, identifies which of the two frequencies is present by finding the strongest FFT bin, and outputs the corresponding binary value.



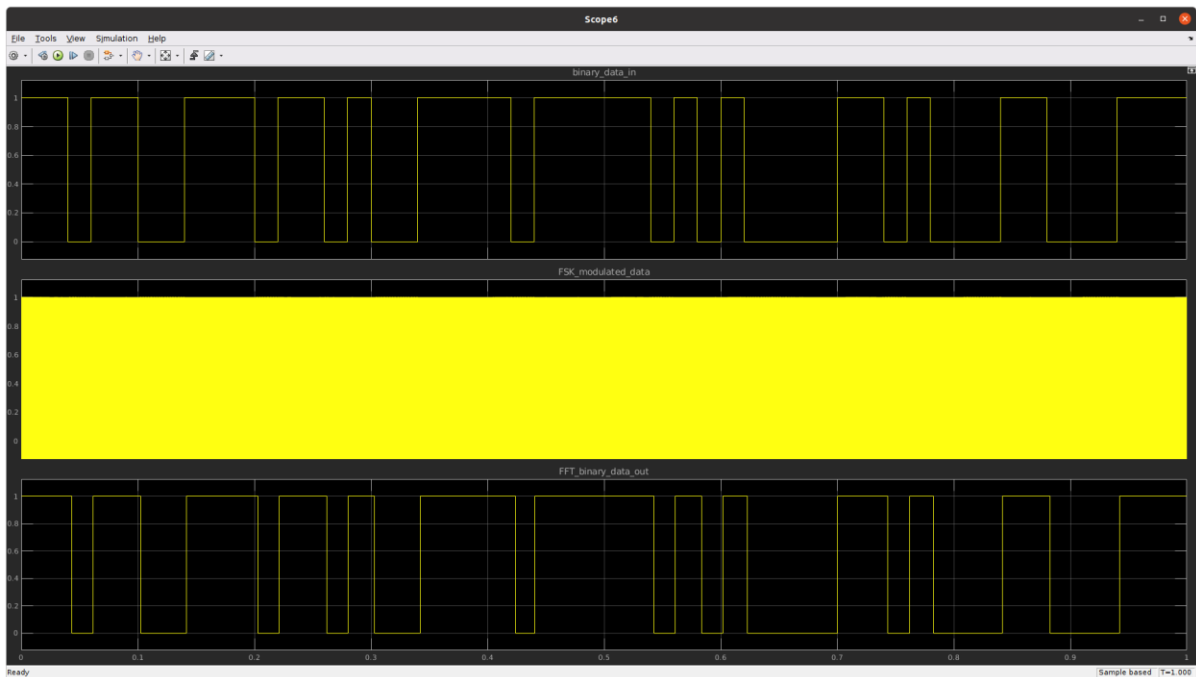




- FSK_modulated_data (Inport)**
 Receives the incoming FSK-modulated signal (time-domain samples).
- Buffer / Frame block (square icon with stacked lines)**
 Groups incoming samples into frames of one symbol duration so the FFT can be applied on a per-symbol basis.
- FFT block**
 Performs the Fast Fourier Transform on each symbol frame to convert it from the time domain to the frequency domain.
- $|FFT|^2$ (Magnitude Squared)**
 Computes the power spectrum by taking the magnitude squared of the FFT output. This shows the energy present at each frequency bin.
- 1/2048 (Gain)**
 Normalizes the FFT power values (for scaling purposes, does not affect detection).
- Find Extrema block**
 Scans the power spectrum and finds the maximum magnitude value and its index (frequency bin where the strongest signal is located).
- Compare block (>) with constant 20**
 Compares the index of the strongest frequency bin against a threshold value (20) to classify the detected symbol:
 - If the index is greater than 20 → logical '1'
 - If the index is less than or equal to 20 → logical '0'

- **binary_data_out_FFT (Output)**

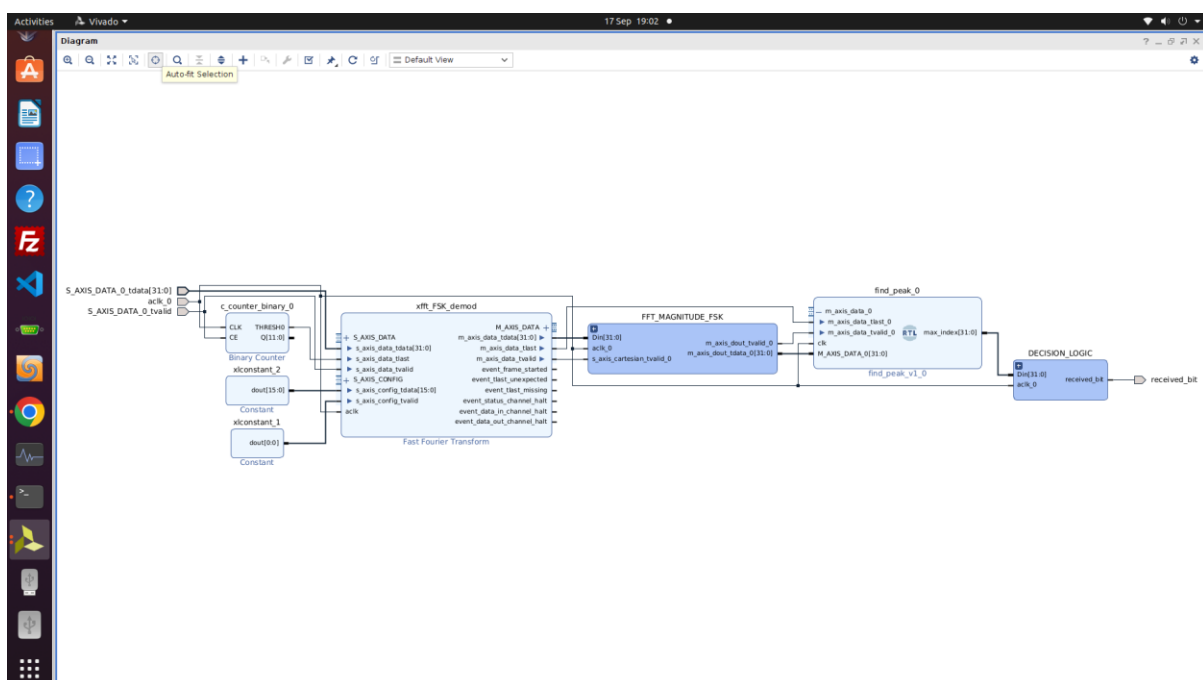
Outputs the detected binary bit corresponding to the frequency present in each symbol period.



The scope plot shows how demodulation is following the input data.

VIVADO FFT BASED FSK DEMODULATION

The FFT based demodulator block design follows:

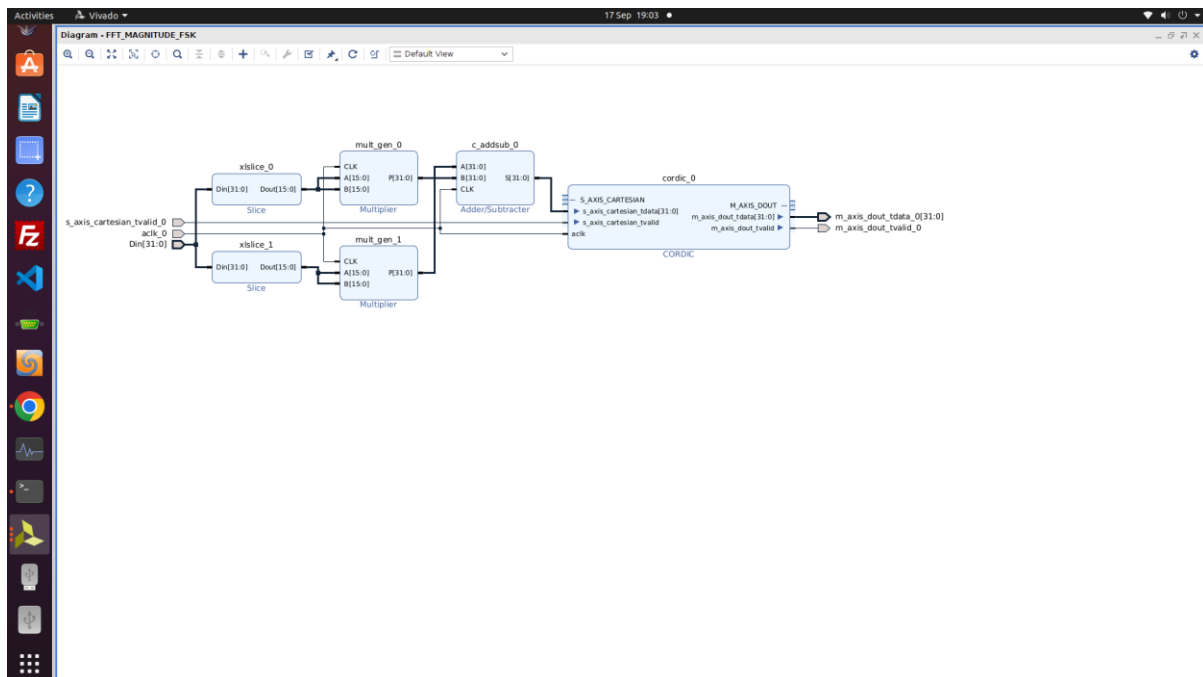


I/Q modulated FSK input data, is applied to the Xilinx FFT IP core.

A counter is used to generate the TLAST signal.

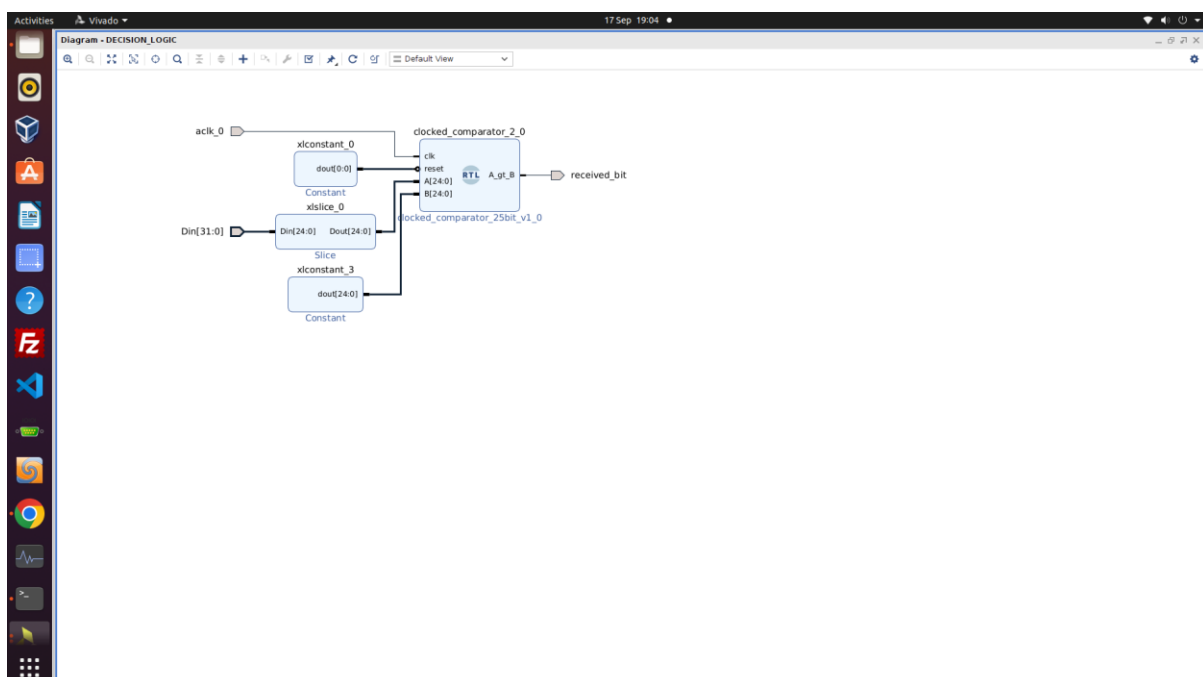
The FFT provides output data in complex form.

The following block computes the magnitude of the elements:



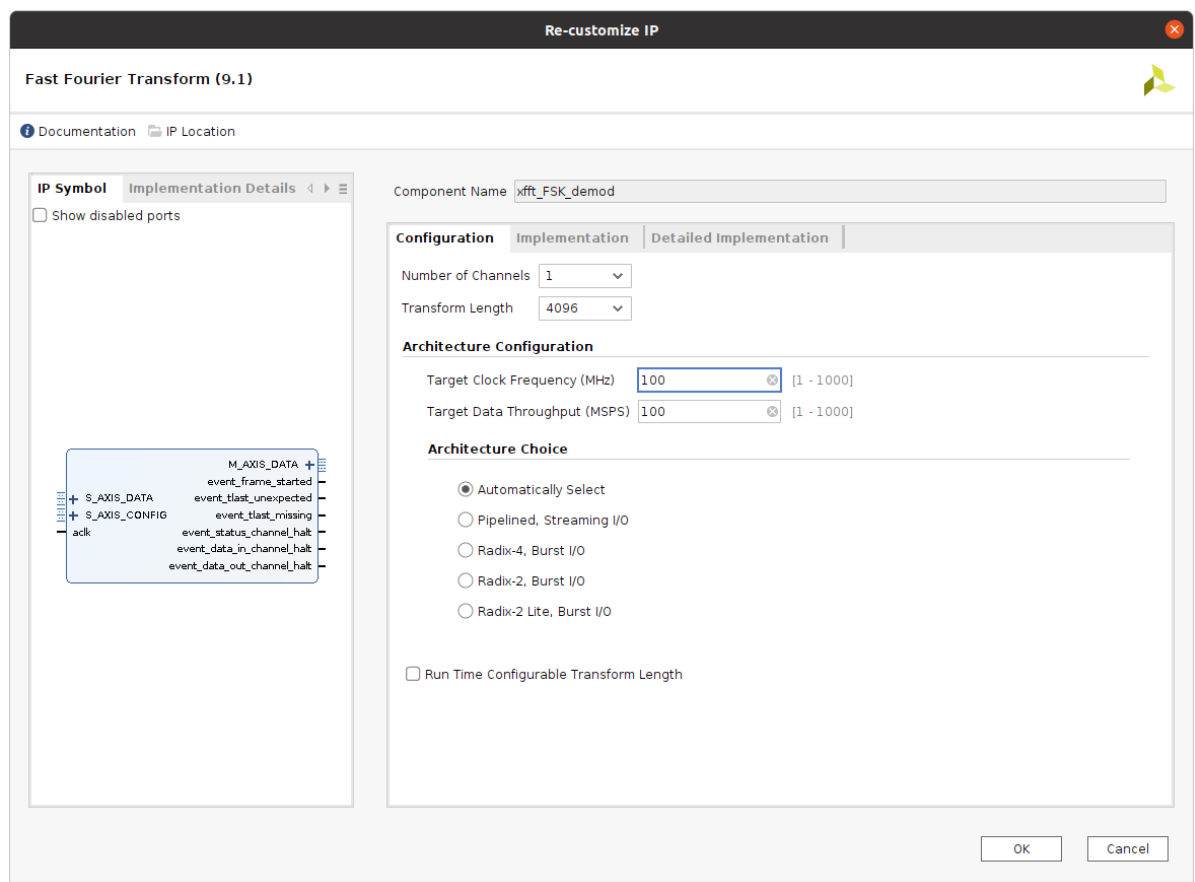
A peak detector is then used to find the index position of the peak amplitude.

The following decision logic:



Is used to establish whether the peak FFT component belongs either to a 0 or to a 1.

Th FFT IP-core is configured as follows:



The CORDIC block is set as:

CORDIC (6.0)

DocumentationIP Location

IP SymbolImplementation Details

Show disabled ports

+ S_AXIS_CARTESIAN
- aclkM_AXIS_DOUT +

Component NameFFT_MAGNITUDE_FSK/cordic_0

Configuration OptionsAXI4 Stream Options

Configuration Parameters

Functional SelectionSquare Root

Architectural ConfigurationParallel

Pipelining ModeMaximum

Data FormatUnsignedFraction

Phase FormatRadians

Input/Output Options

Input Width32 [8 - 48]

Output Width32 [8 - 48]

Round ModeTruncate

Advanced Configuration Parameters

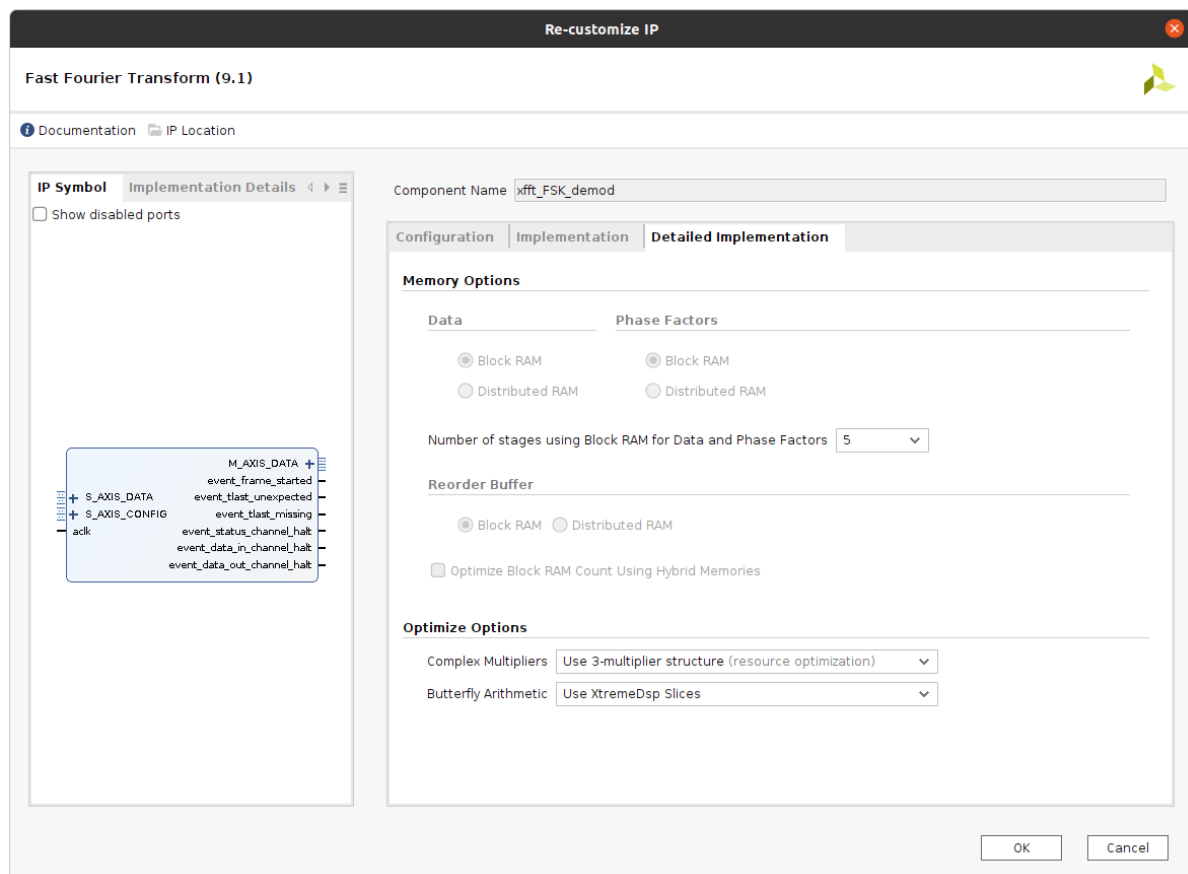
Iterations0 [0 - 48]

Precision0 [0 - 48]

☐ Coarse Rotation

Compensation ScalingNo Scale Compensation

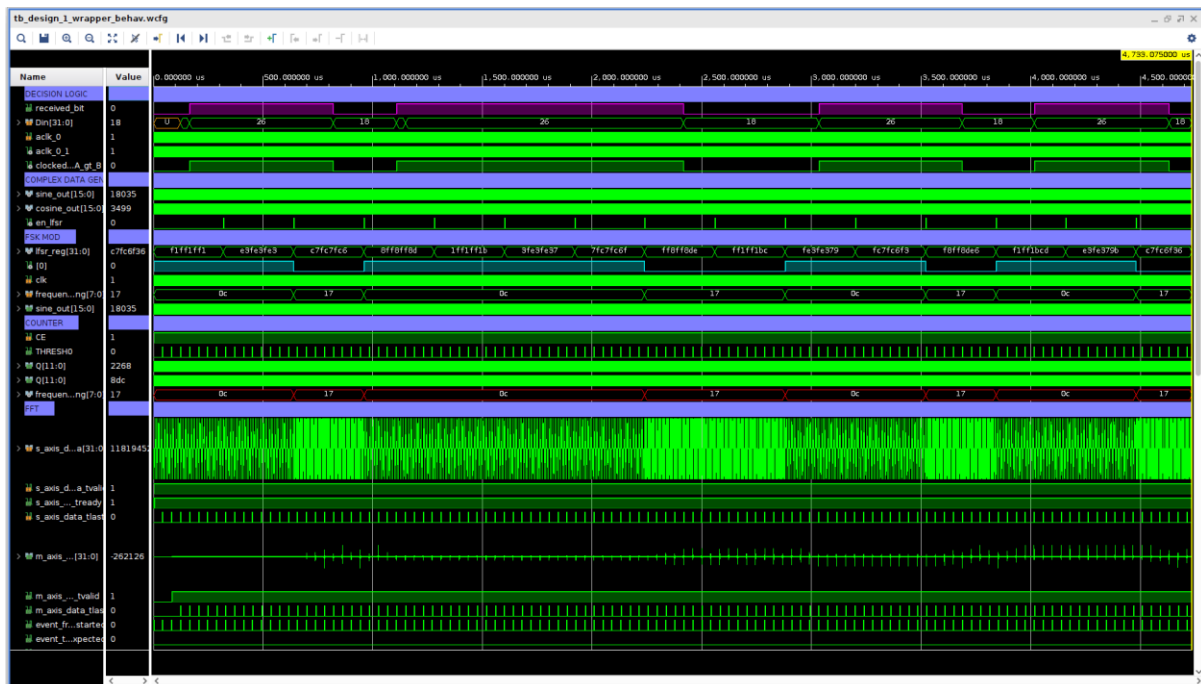
OKCancel



TESTBENCH

The provided testbench verifies the functionality of the design_1_wrapper module by feeding it a simulated binary FSK signal. It uses a Linear Feedback Shift Register (LFSR) to generate a pseudo-random bit sequence, where each bit selects one of two predefined frequencies. These frequencies are provided as input to a I/Q sinewave generator component, which outputs corresponding sine and cosine wave samples (I/Q data). The sine and cosine samples are combined into a 32-bit word and continuously sent to the DUT (design_1_wrapper) through its AXIS-style data interface. A clock generator drives the entire system, and a reset is asserted at the start of the simulation to initialize all logic. By switching the frequency according to the LFSR bits, the testbench produces an FSK-modulated signal that allows testing of the demodulation logic inside the DUT.

By looking at key waveforms:



We can see the demodulated purple signal following the light blue input modulating data generated by the LFSR.

